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SENIOR LEADER PERSPECTIVE

Accelerating with AI: Cyber Defense at Speed, Scope, Scale

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The U.S. Department of War developed its cyber warfare framework during an era in which human decision-making could generally keep pace with technological change and adversarial activity. Today, advances in frontier artificial intelligence (AI) and autonomous cyber capabilities are transforming the need for cyber operational speed, scope, and scale. As traditional timelines for detection, decision-making, and response continue to compress, preserving mission assurance will require more than technological innovation alone. As Commander of the DOW Cyber Defense Command (DCDC) and Director of the Defense Information Systems Agency (DISA), Army Lt. Gen. Paul T. Stanton is leading his organizations to campaign accordingly, setting conditions to effectively engage the enemy through defense. Stanton charges the department with an imperative to continue strengthening and modernizing our cyber architecture, accounting for full-scope resourcing and operationalizing our partnerships with industry.

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Lt. Gen. Paul T. Stanton serves as Commander of the Department of Defense Cyber Defense Command (DCDC) and Director of the Defense Information Systems Agency (DISA). In these roles, he leads the defense, operation, and modernization of the Department of War Information Network (DOWIN) and oversees a global workforce of more than 20,000 military, civilian, and contractor personnel supporting defense communications and command-and-control capabilities worldwide. Prior to assuming his current position, Stanton served as Commanding General of the U.S. Army Cyber Center of Excellence and Fort Eisenhower. A 1995 graduate of the U.S. Military Academy at West Point, he commissioned into the Infantry and transitioned to the Cyber Branch in 2015. His career includes command and leadership assignments in operational, joint, and cyber organizations, including the U.S. Army Cyber Protection Brigade, U.S. Cyber Command, and U.S. Army Cyber Command. Stanton holds degrees from West Point, the University of Illinois, and Johns Hopkins University.



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THE CHANGING CHARACTER OF CYBER WARFARE

As *The Cyber Defense Review* marks another milestone in its contribution to the study and practice of cybersecurity, it is worth reflecting on how much has changed and what those changes reveal about the future of the cyber warfighting domain. Over the past decade, the department has made significant progress in cyber warfare planning, capabilities, and operations. Today, every major national security mission depends upon digital infrastructure, data, software, communications networks, and the organizations that operate them.

In 2018, U.S. Cyber Command's strategy of persistent engagement through a professional force set the stage for cyber operations (Nakasone 2019). The approach has matured into cyber campaigning that remains a sound framework (U.S. Cyber Command PAO 2022; Fischerkeller, Goldman, and Harknett 2025). Building upon this solid foundation, investments in cybersecurity modernization, cloud adoption, zero-trust architecture, cyber workforce development, and enterprise visibility have produced meaningful improvements. These aligned efforts have enhanced our ability to identify risks, understand our networks, and transition our operational mindset from reactive incident response to active, synchronized defense against sophisticated adversaries. However, we must do more.

The DOW Information Network (DOWIN) is a strategic weapon system that underpins global command and control, intelligence, logistics, and military operations. In support of its pivotal role, the department has made significant progress in strengthening its security and resilience. However, many of the systems governing cybersecurity today were established when technological change occurred at a comparatively slower pace. Though they continue to serve important functions, preserving mission assurance now and into the future will require transforming legacy structures to adapt to the speed, scope, and scale of tomorrow's fight. Throughout this transition, we must continue to focus on what matters: effective mission operations calibrated against enemy intent and capability. Our focus must remain on campaigning to deliver exactly what our warfighting commanders require to execute and sustain their missions at the speed of relevance.

The challenge of operating at speed extends far beyond technology. It demands structural transformation toward modern and agile cyber architecture, sustained, full-scope enterprise resourcing, and seamlessly integrated partnerships with the commercial sector. The future of cyber campaigning will not be determined by which organizations possess the most advanced technologies, but by which organizations can most effectively convert awareness into action, coordinate across complex ecosystems, and sustain critical missions under increasingly dynamic conditions.

SHARED RISK EMERGES

We have crossed a strategic threshold. The emergence of frontier AI models, increasingly autonomous cyber systems, and highly scalable digital technologies is accelerating the evolution of cyber warfare. Due to global reliance on cyber systems, frontier AI models introduce a disruptive surge to cyber operations. We are now engaged in a fierce capability competition that will dictate the future character of warfare, intelligence, and national security. Frontier AI models and autonomous systems are compressing timelines across vulnerability discovery, software analysis, exploit development, operational planning, and influence operations—creating a narrow margin between discovery and exploitation. As algorithms become exponentially more capable, traditional buffer zones for reactions are evaporating, requiring automated, near-instantaneous action to outmaneuver adversaries.

This challenge means cyber defenders must access information and rapidly translate it to awareness within a synchronized, operational-level campaign before adversaries can achieve effects. Doing so effectively requires the institutions responsible for defending national security missions to adapt quickly. The imperative for strategic adaptation becomes clear when considering what is at stake. The Joint Force operates with unprecedented connectedness because the department's digital architecture enables the synchronization of global command and control, distribution of real-time intelligence, coordination of complex logistical and sustainment networks, and the projection of strategic deterrence. Our modern operational advantages—speed of decision, synchronized maneuver, and unified command—depend upon the availability and absolute integrity of the DOWIN.

Compounding this complexity, our digital architecture is not a closed system. Joint operations rely on commercial cloud services, civilian telecommunications, and critical infrastructure owned and operated entirely outside the DOW. Mission assurance is therefore inextricably tied to the resilience of the commercial sector. Our adversaries clearly understand this asymmetric vulnerability. They systematically target these commercial providers precisely because their networks support our warfighting

capabilities, exposing private-sector infrastructure to highly capable nation-state intent to disrupt our decision-making cycles and degrade our combat readiness.

Operating without comprehensive visibility creates severe institutional blind spots on both sides of the civil-military relationship. Commercial technology developers possess unparalleled expertise in AI architecture, but they frequently lack the warfighting context needed to determine whether a theoretical vulnerability poses a mission risk. Conversely, the defense establishment understands the operational landscape but depends on industry-designed systems and engineering expertise. This separation makes it difficult for defenders to distinguish between network noise and unprecedented, machine-generated attacks. Decisions regarding cyber operations must account for this shared risk environment, ensuring that our institutions, partnerships, and operating concepts evolve at a sufficient pace to preserve military effectiveness against machine-speed threats.

ADAPTATION AS A STRATEGIC IMPERATIVE

While the enduring principles that underpin military effectiveness remain constant, the application of those principles must evolve to match a shifting operational environment. Today, frontier AI models are driving significant and rapid evolution, placing an unprecedented premium on our institutional capacity to adapt. This adaptation requires a parallel evolution of our technical architecture, our institutional frameworks, and our operational partnerships. This demands we align our doctrine, command structures, training systems, and administrative policies with the speed of modern conflict.

We must systematically reassess how we measure risk, how we leverage authorities, how decentralized forces coordinate, and how mission priorities dictate defensive actions. Mission owners, network defenders, critical infrastructure operators, and commercial partners will shape operational outcomes through their collective decisions. Preserving mission assurance requires agile digital architecture operated by fully resourced institutions that partner with industry to defend our nation.

Designing an Architecture for Agility

To outpace our adversaries in an environment defined by automated speed, we cannot simply accumulate software or layer new tools on top of fragile systems. We must move faster by designing and implementing the resilient environment we need. While we are making progress, we must accelerate our transition by systematically eliminating technical debt, replacing obsolete infrastructure and establishing a secure, standardized enterprise baseline across the force. This enduring digital foundation is the prerequisite for processing data at the speed of conflict and enabling seamless command and control across every warfighting domain.

Well-designed and modernized architecture serves as the critical enabler to achieve operational velocity. It establishes the necessary structural clarity and real-time operational context across our vast, globally distributed enterprise. The architectural framework empowers decentralized commanders to make decisions that are locally risk-informed, yet globally risk-bounded. This operational standard is enforced through a zero-trust architecture that integrates federated identity, credential, and access management with automated, context-aware policy enforcement. By enforcing cryptographic boundaries and dynamic network segmentation at the tactical edge, we can isolate and contain localized compromises before they propagate across our global transport layers. Automating this

containment enables tactical commanders to configure and maneuver their networks for immediate combat requirements while structurally preventing their localized decisions from introducing systemic vulnerabilities into the broader enterprise.

Delivering this agile architecture requires overhauling our acquisition models and administrative processes. We must cultivate a responsive delivery ecosystem that leverages rapid prototyping and iterative commercial development in close coordination with our industry partners. To achieve operational speed at scale, the department must prioritize foundational enterprise transformation over isolated technological novelties. Investing in premature or fragmented AI initiatives at the expense of cohesive architecture is strategically counterproductive: standardized, well-designed architecture is the prerequisite for fully leveraging mature AI capabilities. Slow is smooth and smooth is fast.

Full DOTmLPF-P Resourcing

Effectively running a modernized architecture and operationally controlling that architecture requires comprehensive institutional resourcing. We cannot move at increased and sustained speed, scope, and scale with solutions that only address parts of the problem. We must align our joint planning using the DOTmLPF-P doctrine, organization, training, materiel, leadership and education, personnel, facilities, and policy framework to structurally synchronize our capabilities (Joint Chiefs of Staff 2026).

Starting with personnel, cyber warriors must profoundly understand the mission they support, the dependencies that enable those missions and the operational consequences of cyber risk. They must be prepared to operate tactically alongside highly capable and automated systems while retaining responsibility for strategic and operational consequences. Ultimately, our decisive advantage in cyberspace will not be the algorithms we deploy, but the ingenuity, adaptability and operational mindset of the professionals who wield them.

Developing our professionals will require processes that can keep pace with updated tactics, techniques, and procedures. We need trained, accountable personnel who follow standardized doctrine that ensures disciplined execution to run architecture with the full power of AI capabilities. We cannot coordinate action without a shared understanding of how we will execute. Designing and modernizing our digital environment is useless if we fail to qualify the professionals who wield it. Adhering to this comprehensive planning construct provides the essential blueprint for full-spectrum force resourcing, moving us beyond a compliance-driven model to a culture of rigorous, periodic qualification on our cyber systems. We must build readiness.

This readiness is crucial when confronting the staggering speed of frontier AI, which collapses the window for vulnerability discovery to active exploit from months to minutes. In this high-velocity threat environment, passive security updates and check-the-box patching are obsolete. We must enforce a rigorous standard of action plus validation, confirming that every configuration change and security remediation actually closes the vulnerability without disrupting our broader operational capabilities. Sustaining mission-critical operations under these contested conditions requires us to view cyber defense not as traditional technology procurement, but as a core warfighting capability fully aligned with the long-term training, testing, and readiness requirements embedded in the DOTMLPF-P framework, including our industry partners.

Operationalizing Partnerships with Industry

We can no longer afford to assess the shared risk environment through an isolated lens. The digital ecosystem that supports national security missions extends beyond the boundaries of the DOW. This reality reflects both the innovation and scale that commercial industry brings to national defense and the deeply interconnected nature of modern operations. Consequently, the resilience of national security missions depends on the resilience of the broader ecosystem that enables military power.

Interdependence with industry is not new, but the pace of cyber competition and contention is changing its implications. Transactional information sharing between government and industry remains valuable but is no longer sufficient. As timelines compress, it is the ability to act collectively on shared information that matters. Government, industry and critical infrastructure operators must develop shared understanding, assess mission consequences and coordinate defensive actions quickly enough to influence outcomes. The objective is not simply greater awareness—it is reducing the time required to move from awareness to coordinated response.

Operational integration at speed requires trusted relationships, shared planning, routine exercises and clearly understood responsibilities well before a crisis occurs. Organizations that attempt to establish those relationships during an incident are unlikely to operate at the speed future conditions will require. The goal is to move beyond the transactional, bilateral nature of service-level contracts and programmatic task orders toward deep, operationalized teamwork. The department and its industry partners can best achieve integration through coordinated incident response, rapid vulnerability mitigation and proactive defenses. Cyber defense in the frontier AI era relies on integrated teammates ready to campaign together with synchronized defensive actions against common adversaries to achieve a level of collective resilience that contracts alone can never guarantee.

CONCLUSION

The true measure of our future success will not be found in isolated pockets of excellence, but in our capacity to proactively campaign for rapid, collective action. In a domain where milliseconds matter, the artificial boundaries between military organizations, government agencies and commercial partners must give way to a seamless, integrated defense. When we align our mission purpose, share our visibility, and synchronize our responses, we transform our interconnectedness from a potential vulnerability into our greatest asymmetric advantage. It is this unified effort—orchestrated with unprecedented efficiency and mutual trust—that will allow us to outpace any adversary.

Defending the DOWIN requires more than cutting-edge technology—it demands an operational mindset rooted in resilient architecture, fully resourced institutions and commercial teamwork to operate with synchronization. Ultimately, the decisive advantage in the future of cyberspace will not belong to the organization that possesses the most advanced algorithms or the most expansive networks. It will belong to the organization that is ready and can adapt the fastest to empower its cyber professionals to execute at the speed, scope, and scale of the fight. For the DOW, adapting to this new reality is not just an administrative goal—it is a strategic and operational imperative. If we are to secure the systems that enable military power and preserve our ability to fight and win in a contested environment, we must build an enterprise designed not just to withstand modern cyber conflict, but to dictate it.

REFERENCES

Fischerkeller, Michael P., Emily O. Goldman, and Richard J. Harknett. 2025. "Setting the Stage: Cyber Contingency Campaigning." *Lawfare* (August 28, 2025). https://www.lawfaremedia.org/article/setting-the-stage--cyber-contingency-campaigning.

Joint Chiefs of Staff. 2026. *Charter of the Joint Requirements Oversight Council and the Joint Force Requirements Process.* CJCSI 5123.01J. https://www.jcs.mil/Portals/36/Documents/Library/Instructions/CJCSI%205123.01J.pdf.

Nakasone, Paul M. 2019. "A Cyber Force for Persistent Operations." *Joint Force Quarterly,* no. 92, 10–14. https://ndupress.ndu.edu/Media/News/News-Article-View/Article/1736950/a-cyber-force-for-persistentoperations/.

U.S. Cyber Command PAO. 2022. *CYBER 101 – Defend Forward and Persistent Engagement,* October 25, 2022. https://www.cybercom.mil/Media/News/Article/3198878/cyber-101-defend-forward-and-persistent-engagement.